

Xinran Yu

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Education

University of Illinois Urbana–Champaign Ph.D. in Mathematics, Advisor: Prof. Pierre Albin M.S. in Mathematics, May 2019	May 2026
University of Liverpool & Xi'an Jiaotong–Liverpool University B.Sc. in Mathematics with Honors & B.Sc. in Applied Mathematics (dual degree, 2+2 program) Thesis: <i>Analytic Continuation and Riemann Surfaces</i> Advisor: Dr. Jon Woolf	June 2017

Research Interests

Geometric Analysis, Conformal Geometry, Mathematical Relativity, Witten Complexes on Singular Spaces.

Preprints and Publications

1. Spectral asymmetry, supersymmetry and the equivariant Riemann-Roch defect
with Gayana Jayasinghe and Alex R. Taylor, [arXiv:2603.26057](#) 2026
2. Conformally Compact Metrics and the Lovelock Tensors 2025
[arXiv:2505.24188](#), submitted
3. Witten Instanton Complex and Morse-Bott Inequalities on Stratified Pseudomanifolds 2024
with Gayana Jayasinghe and Hadrian Quan, [arXiv:2412.12003](#), submitted

Talks

AMS Fall Central Sectional Meeting, Saint Louis University (Invited) “Asymptotic Behavior of Conformally Compact Lovelock Metrics”	Oct 2025
Geometric Analysis Seminar, UIUC (Invited) “Fefferman-Graham expansion of Conformally Compact Lovelock Metrics”	Sep 2025
Great Lakes Mathematical Physics Meeting (GLaMP), University of Kentucky “Asymptotic Behavior of Conformally Compact Lovelock Metrics”	Jun 2025
Graduate Student Geometry and Topology Seminar, UIUC “Dirichlet-to-Neumann Map for Conformally Compact Einstein Metrics”	Oct 2024
AWM Graduate Student Colloquium, UIUC “Conformal Geometry in Lovelock Gravity Theories”	Apr 2024

Honors & Awards

Boya Postdoctoral Fellowship	2026
Susan C. Morisato IML Scholarship, UIUC	Summer 2025
Ruth V. Shaff & Genevieve I. Andrews Fellowship, UIUC	Spring 2024
Teachers Ranked as Excellent by their Students, UIUC	2021
Wills Prize in Mathematics, UoL	2017
IMA Prize, UoL	2017
University Academic Achievement Award, XJTLU	2014–2015

Teaching Experience

At UIUC

2019-2026

Instructor

- MATH 231: Calculus II (Spring 2025, Spring 2022)

Teaching Assistant

- MATH 241: Calculus III (Spring 2026, Fall 2021). Teachers Ranked as Excellent
- MATH 220: Calculus (Fall 2024)
- MATH 231: Calculus II (Spring 2021). Teachers Ranked as Excellent

Grader

Graduate Courses

- MATH 518–519: Differentiable Manifolds I–II (Fall 2023, Spring 2023)
- MATH 553: Introduction to Partial Differential Equations (Spring 2023)
- MATH 541: Functional Analysis (Spring 2023)
- MATH 512–514: Algebraic/Complex Algebraic Geometry (Fall 2022)
- MATH 525: Algebraic Topology I (Spring 2021)

Undergraduate Courses

- MATH 416: Abstract Linear Algebra (Fall 2025, Fall 2022)
- MATH 402, 403: Non-/Euclidean Geometry (Fall 2023)
- MATH 423: Differential Geometry (Fall 2023)
- MATH 286: Introduction to Differential Equations Plus (Fall 2020)
- MATH 489: Dynamics and Differential Equations (Fall 2019)
- MATH 417, 418: Introduction to Abstract Algebra I (Spring 2019), II (Fall 2019)

Mentorship

At UIUC

Illinois Mathematics Lab, Mentor

2024–2025

Projects conducted in collaboration with the Mathematics Library

- *Historical Mathematical Models* (Spring 2025) — Supervised 5 undergraduates.
 - *Documenting Historical Mathematical Models* (Fall 2024) — Supervised 4 undergraduates.
- Project received the Nancy D. Anderson Undergraduate Research Award.

Illinois Geometry Lab, Mentor

Fall 2019

- *Simulating Multi-Soliton Solutions to NLS and KdV* — Supervised 4 undergraduates.

Services & Enrichment

At UIUC

German Translation Project, Mathematics Library

2024–2025

International TA Panelist

Aug 2022

At UoL

Staff-Student Liaison Committee

2016–2017